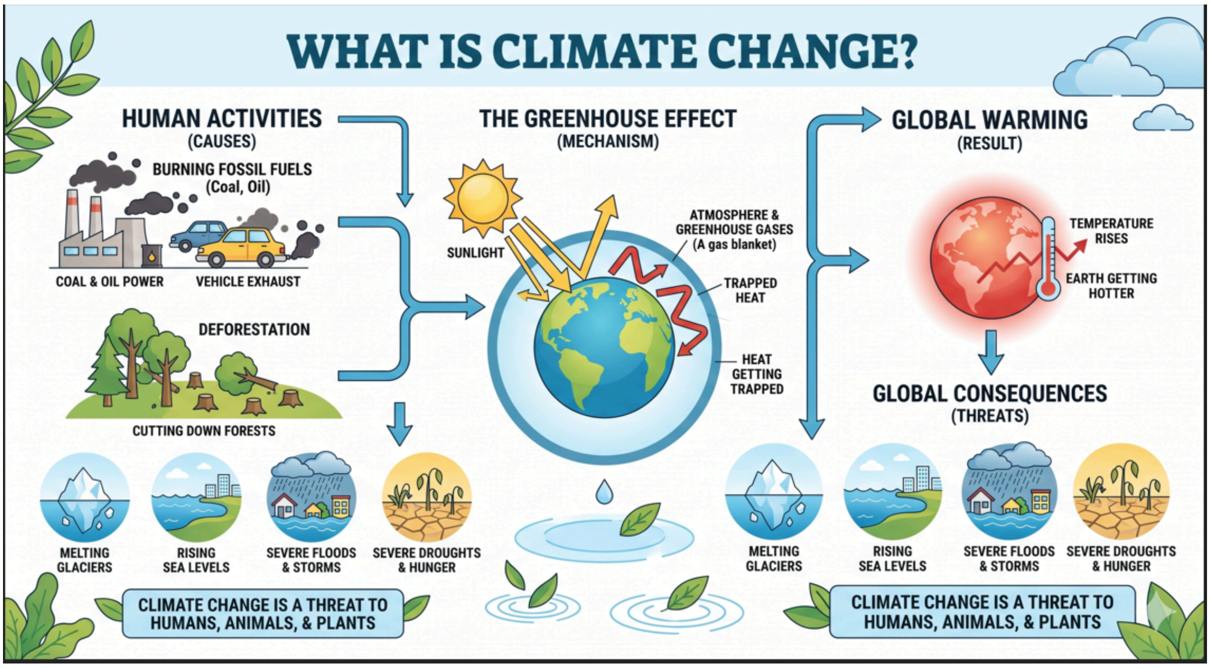
# Climate change and environmental degradation
# Environmental Problems and Sustainable development

**Climate change: -**

Climate change refers to those unusual changes in the Earth's average temperature and long-term weather patterns that are disrupting the planet's natural balance. Its primary cause is human activity, such as the burning of fuels (coal and oil) in factories and vehicles and deforestation, which increases harmful gases in the atmosphere and traps the sun's heat around the Earth. Due to this continuously increasing heat or 'global warming,' glaciers are melting rapidly, sea levels are rising, and natural disasters such as untimely rains, severe floods, and droughts are increasing worldwide. In simple words, this is the deterioration of the Earth's environment that has become a massive threat to the lives of humans, animals, and plants alike.

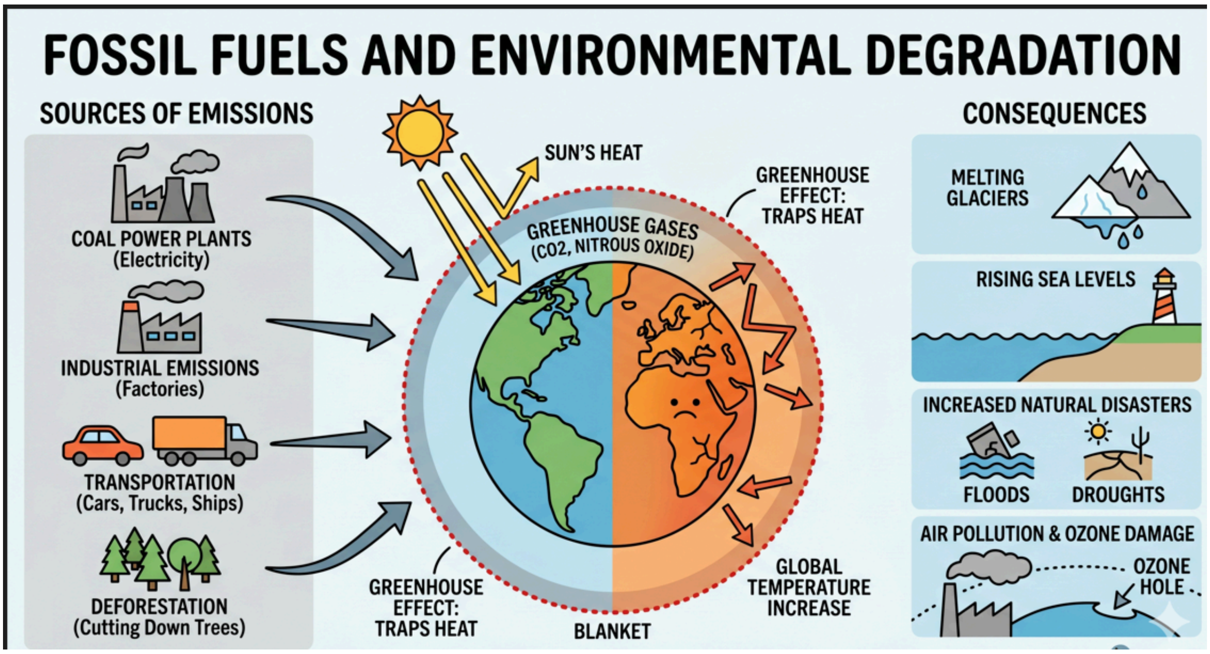



**Reasons behind Climate change: -**

## 1. Role of Fossil Fuels, Industrial Emissions, and Transport

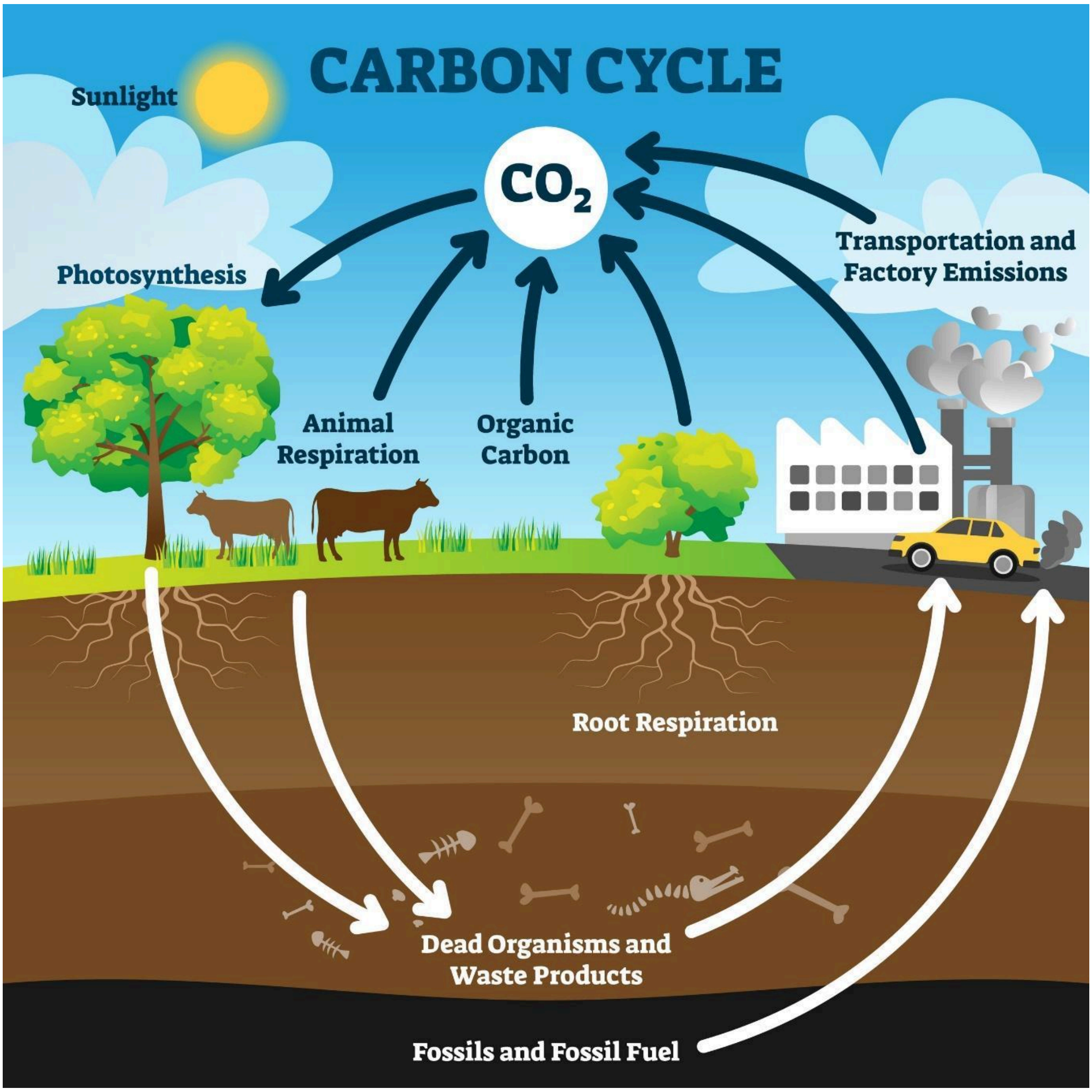
The biggest and most fundamental cause of global temperature rise and environmental degradation is the excessive use of **fossil fuels**, namely coal, oil, and gas. When we burn these fuels for electricity production, running factories, or for vehicles, there is a large-scale emission of greenhouse gases such as carbon dioxide (CO2) and nitrous oxide into the atmosphere.
These gases wrap around the Earth like a blanket and prevent the sun's heat from escaping back into space, which is known as the **'greenhouse effect.'** Industrial activities not only make the atmosphere toxic through smoke, but the gases produced during chemical processes also damage the ozone layer. The transport sector, which is entirely dependent on petroleum products, is constantly increasing air pollution, which is not only melting

glaciers but also causing a dangerous rise in sea levels.




## Impacts of Deforestation, Agriculture, and Waste Accumulation

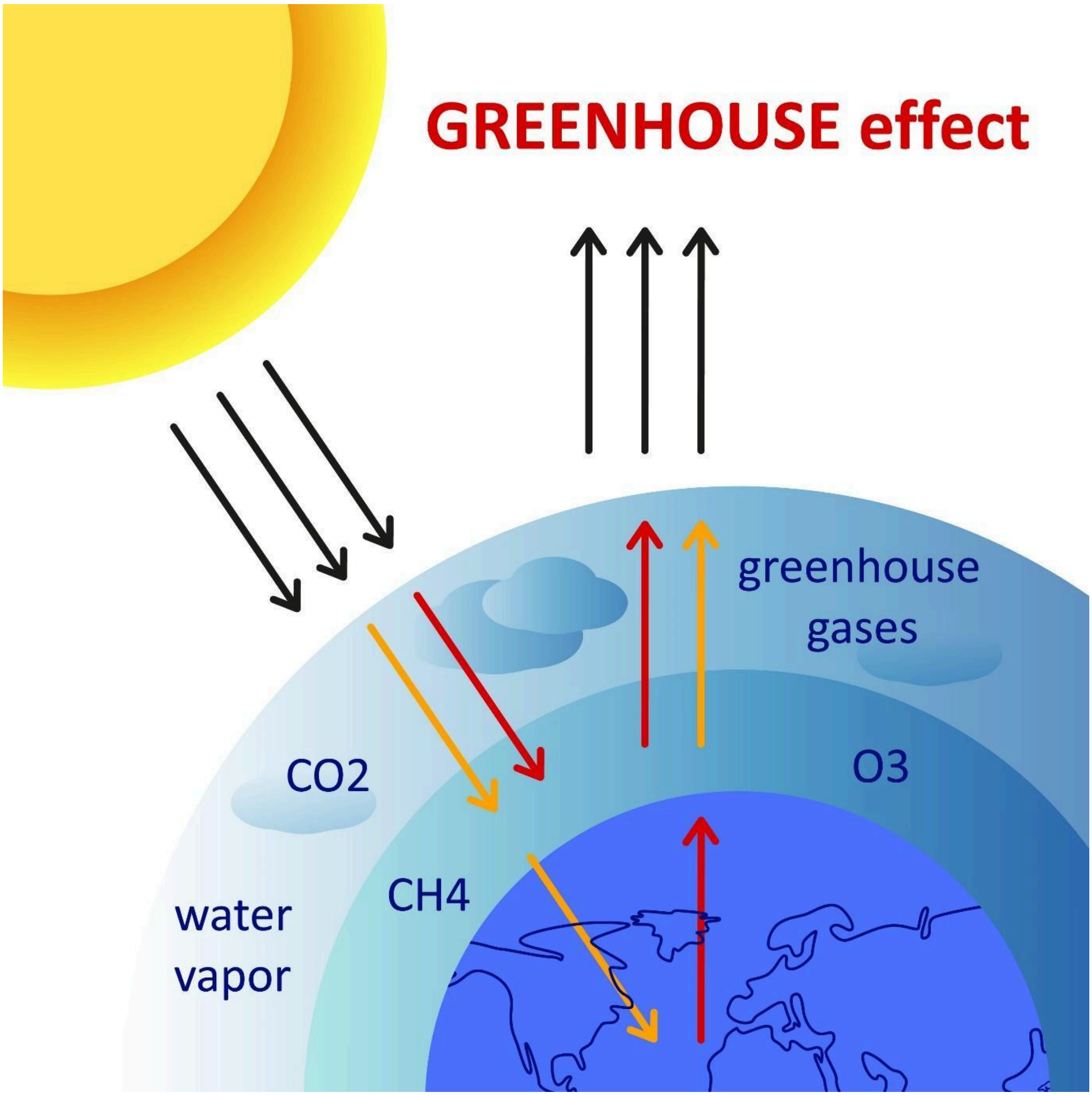
On the other hand, human interference has severely damaged the Earth's natural defense system—the forests. Trees naturally keep the environment cool by absorbing carbon dioxide, but when forests are cleared for housing schemes or timber, the process of carbon absorption stops. Furthermore, the carbon stored in these trees is released back into the atmosphere, increasing the heat.




Along with this, agriculture and livestock farming are also major sources of methane (CH4) emissions. Specifically, the digestive systems of grazing animals and the use of fertilizers produce gases that have a much higher capacity to trap heat compared to carbon dioxide.
Furthermore, when large piles of garbage (landfills) near cities decompose, they also emit methane gas. This contributes to the increasing intensity of climate change, leading to unusual floods and severe droughts.

## The Reality of Greenhouse Gases and Their Impact on the Environment
## Important Gases and the Heat-Trapping Process

Greenhouse gases actually act as a natural shield around the Earth, but an artificial increase in their quantity has become a menace for the planet. **Carbon dioxide (CO2)** is the most impactful gas on this list; it enters the atmosphere from the burning of coal, oil, and gas and remains there for centuries, keeping the sun's heat trapped.




Along with it, the role of **methane ($CH_4$)** is also extremely dangerous; although it spends less time in the atmosphere compared to carbon, its heat-absorbing capacity can be approximately **25 to 80 times higher** than carbon dioxide. This gas is mostly emitted from the digestive systems of livestock, rice paddies, and landfills.

In this natural process, **water vapor** plays a unique role; as the atmosphere warms, the amount of vapor increases, which absorbs more heat and gives rise to a never-ending cycle (**feedback loop**) of heat.

## Effects of Climate change on Pakistan

Pakistan is among the most vulnerable countries to climate change, even though it contributes less than 1% to global carbon emissions. The glaciers in the northern regions, which are Pakistan's largest source of fresh water, are melting rapidly. This process is causing Glacial Lake Outburst Floods (GLOFs) on one hand, while on the other, it creates a long-term risk of decreasing water levels in rivers, posing a serious threat to the country's water security.

Agriculture is the backbone of Pakistan's economy, but climate change is severely paralyzing this sector. Untimely and unusually heavy rains, prolonged droughts, and intense heatwaves have disrupted the natural cycles of sowing and harvesting. Major crops like wheat, cotton, and rice are increasingly being destroyed by either floods or water shortages, leading not only to economic hardship for farmers but also to rising food insecurity and inflation across the nation.

Extreme heat and natural disasters are directly impacting human health and social stability. In the plains of Sindh and South Punjab, temperatures often exceed 50°C, significantly increasing the risk of heatstroke and other heat-related illnesses. Events like the devastating 2022 floods have displaced millions, turning them into "climate refugees," while stagnant water has led to the rapid spread of diseases such as malaria, dengue, and cholera, placing an enormous burden on the healthcare system.

**Climate change is having a big impact on how plants live and grow.**

**1. Effects on Temperature and Growth**
Rising temperatures are messing up the natural way plants grow. When it gets too hot, plants get stressed and stop growing to save their energy. Here are three examples: first, many trees and plants grow flowers way too early, which confuses their life cycle; second, strong heatwaves make plants dry up and burn their leaves; and third, some types of plants are disappearing from their original homes and can now only survive in cool, high mountains.
**2. Water and Harsh Weather**
Climate change makes it hard to know when it will rain, which is very dangerous for plants. Plants suffer if they get too much or too little water. For example: first, long times without rain (droughts) dry out the soil and kill the plants; second, sudden floods cover plant roots and take away their oxygen, causing them to rot; and third, as the sea level rises, the soil near the coast becomes too salty, which kills the local plants.
**3. Pest Attacks and Less Nutrition**
Changes in the weather make plants weak and make it harder for them to fight off diseases. When the environment changes, plants can't defend themselves against hungry bugs. For example: first, because winters are warmer, bad bugs don't die off and instead attack large farms; second, having too much $CO_2$ gas in the air makes plants grow faster, but they end up having less protein and minerals; and third, flowers might bloom at a different time than when the bees or birds come to visit, which stops new seeds from growing.

## Smart crops

Scientists are currently working on developing "smart crops" that can not only withstand the changing climate but also maintain their nutritional value. For this purpose, work is being carried out on the following three major methods:
**1. Biofortification (Nutritional Enhancement)**
Since the excess of carbon dioxide ($CO_2$) in the atmosphere is reducing the levels of zinc and iron in crops, scientists are using **biofortification** to develop seeds that naturally contain higher amounts of these minerals than regular crops.
In this process, a plant's ability to absorb minerals from the soil is enhanced through traditional breeding and modern technology. For example, zinc-rich wheat varieties have been introduced in Pakistan and India that are far more beneficial for human health than ordinary wheat.

**2. Resistance Against Drought and Heat**
To protect against water scarcity and sudden heatwaves caused by climate change, crops are being developed that can survive with less water.
The roots of these plants are deeper, allowing them to absorb moisture from the lower layers of the soil. Additionally, **"short-duration crops"** are being created that complete their growth cycle and are ready for harvest before the peak of a severe heatwave. This minimizes losses for farmers and ensures a steady food supply.

**3. Genetic Engineering and CRISPR Technology**
Modern science is now making specific changes to the DNA of plants so they can independently fight against saline (salty) water, pest attacks, and diseases.
Using advanced technology scientists can modify only the specific parts of a plant’s genetic code that make it vulnerable. A prime example is the development of rice

varieties for coastal areas affected by rising sea levels; these varieties can provide excellent yields even in salty soil.

## Sustainable development

**Planting Billions of Trees**
Pakistan is working very hard to protect our Earth by planting billions of trees across the country. Trees are like the lungs of our planet because they breathe in the bad air that causes global warming and give us fresh oxygen to breathe. The government started a big project called the "Ten Billion Tree Tsunami" to turn dry lands into green forests. These new forests help keep the weather cool, bring more rain, and provide safe homes for beautiful birds and animals. By planting more trees, Pakistan is making sure that our future is green, healthy, and full of life.

**Using Clean Energy from the Sun and Wind**
To stop the air from getting dirty, Pakistan is moving away from burning coal and oil, which create a lot of smoke. Instead, we are now using the power of the sun and the wind to make electricity. This is called "Clean Energy" because it does not hurt the environment. Many schools, homes, and even big factories are putting solar panels on their roofs to catch sunlight and turn it into power. In windy places, big wind turbines are being set up to make electricity. Using the sun and wind helps keep our sky blue and our air clean for everyone to breathe.

**Saving Water and Smart Farming**
Water is very precious in Pakistan, so we are learning new and better ways to save every drop. Because the weather is getting hotter and it rains at the wrong times, farmers are being taught "Smart Farming." Instead of flooding the whole field with water, they use special pipes called "Drip Irrigation" that give water directly to the roots of the plants. This saves a lot of water for the future. We are also building big walls and systems in the mountains to protect people from floods when the ice on the mountains melts too fast. This helps our farmers grow food safely even when the climate changes.

**Clean Transport and Green Cities**
Pakistan is also trying to make our cities cleaner by using electric cars, bikes, and buses. These vehicles run on batteries and do not puff out smelly, black smoke like old cars do. This helps to stop "Smog" and keeps the city air fresh. We are also trying to use less plastic and manage our trash better so that it does not end up in our beautiful rivers and oceans. By using green buses and recycling our waste, we are helping to stop climate change. When we keep our cities clean and green, we make Pakistan a much better and happier place for all children to grow up.